Lecture 17

Vacua and Goldstone modes

The curvature of a potential at equilibrium gives an oscillator's squared frequency. For a field, the same curvature gives squared masses. A continuous symmetry can force some of these curvatures to vanish. We identify the directions it forces and distinguish them from other zero curvatures.

Objectives. *We prove the orbit–Hessian Goldstone lemma for a compact group acting orthogonally on real scalar fields. We compute the radial and phase masses of the global $O(2)$ model with the course normalisation.*

By the end of this lecture you will be able to:

- Find the subgroup preserving a chosen vacuum and the orbit dimension.
- Prove that orbit tangents lie in the vacuum Hessian's kernel.
- Count symmetry-forced massless modes and identify accidental zeros.

17.1 Vacua and the linearised field

Let $V : \mathbb{R}^N \to \mathbb{R}$ be C^2 and attain its minimum value $V_{\min}$.

Definition 17.1 (Vacuum and vacuum set)**.** A *vacuum* is a constant field $\phi(x) = u_0$ with $V(u_0) = V_{\min}$. The *vacuum set* is

$$\mathcal{V} = \{u \in \mathbb{R}^N : V(u) = V_{\min}\}.$$

When this set is a smooth submanifold, it is called the *vacuum manifold*.

The set of minima need not be a manifold: $V(u_1, u_2) = u_1^2 u_2^2$ has two crossing axes as its minima. Each compact group orbit considered below is a smooth submanifold of $\mathbb{R}^N$ contained in the vacuum set. As in Lecture 15, a nonzero constant background on $\mathbb{R}^d$ need not belong to L^2. The perturbation, rather than the background, supplies the linear evolution data.

Use the canonical real-scalar density and equation of Lecture 1:

$$L = \frac{1}{2}\sum_{a=1}^{N} \partial_\mu \phi^a \partial^\mu \phi^a - V(\phi), \qquad \Box\phi + \nabla V(\phi) = 0. \tag{17.1}$$

Fix a vacuum $u_0 \in \mathcal{V}$. Its gradient vanishes. Differentiating this equation at $\phi = u_0$ in the direction h gives the *linearised equation*

$$\Box h + \mathsf{M} h = 0, \qquad \mathsf{M} = D^2 V(u_0). \tag{17.2}$$

Definition 17.2 (Mass matrix and massless mode)**.** The Hessian M in (17.2) is the *mass matrix*, whose eigenvalues are the squared masses. An eigenvector with eigenvalue zero is a *massless mode* of the linearised field. Modes are counted with their linear multiplicities.

The matrix is real symmetric. For every $w \in \mathbb{R}^N$, the second derivative of $s \mapsto V(u_0 + sw)$ at zero is $w^{\mathsf{T}}\mathsf{M}w \geq 0$. Thus its eigenvalues are $m_a^2 \geq 0$. An orthogonal change of field components diagonalises the equation into $(\Box + m_a^2)h_a = 0$. The Fourier frequencies are $\nu_a(k) = \sqrt{|k|^2 + m_a^2}$. These equations have the energy-class evolution of Lecture 4. The number of massless components is therefore $\dim \ker \mathsf{M}$.

17.2 The symmetry orbit and its Hessian

Let G be a compact finite-dimensional Lie group and let $\mathcal{R} : G \to O(N)$ be a smooth group homomorphism. It defines the linear action $u \mapsto \mathcal{R}(g)u$. Assume that

$$V(\mathcal{R}(g)u) = V(u) \qquad (g \in G,\ u \in \mathbb{R}^N). \tag{17.3}$$

Orthogonality also preserves the kinetic term of (17.1), so this is a global internal symmetry of the field theory.

Definition 17.3 (Unbroken subgroup and orbit directions)**.** The *unbroken subgroup* at u_0 is its stabiliser

$$H = \{g \in G : \mathcal{R}(g)u_0 = u_0\}.$$

The *vacuum orbit* is $\mathcal{O}_{u_0} = \{\mathcal{R}(g)u_0 : g \in G\}$. For X in the Lie algebra $\mathfrak{g} = T_eG$, let $T_X = d\mathcal{R}_e(X)$. Its orbit direction is T_Xu_0. The continuous direction is *unbroken* when $T_Xu_0 = 0$ and *broken* otherwise.

Differentiating $\mathcal{R}(g)^{\mathsf{T}}\mathcal{R}(g) = I$ at the identity gives $T_X^{\mathsf{T}} = -T_X$. The stabiliser is closed, since it is the inverse image of u_0 under the continuous orbit map. Recall the closed-subgroup quotient theorem from *Topology and Geometry*, Part I, Lecture 8: a closed subgroup is a Lie subgroup, and G/H is a smooth manifold of dimension $\dim G - \dim H$. The action need not be faithful; elements acting trivially belong to H. We also recall from Part I, Lecture 9, that $\exp(sX)$ is the one-parameter subgroup with initial tangent X. Differentiating its representation gives the matrix equation $Y' = T_XY$, hence $\mathcal{R}(\exp(sX)) = \exp(sT_X)$ by uniqueness of that linear ODE.

Lemma 17.4 (Orbit–Hessian Goldstone lemma)**.** *Under these hypotheses, the orbit is a smooth embedded submanifold $\mathcal{O}_{u_0} \cong G/H$ contained in $\mathcal{V}$, with*

$$\begin{aligned} T_{u_0}\mathcal{O}_{u_0} &= \{T_Xu_0 : X \in \mathfrak{g}\} \subset \ker D^2V(u_0), \\ \dim T_{u_0}\mathcal{O}_{u_0} &= \dim G - \dim H. \end{aligned} \tag{17.4}$$

Consequently the linearised field has at least $\dim G - \dim H$ massless modes. It has exactly that many if and only if the Hessian kernel equals the orbit tangent space.

Proof. We first identify the tangent space and its dimension. The map $G/H \to \mathbb{R}^N$ given by $gH \mapsto \mathcal{R}(g)u_0$ is well-defined and injective: two group elements have the same image exactly when they differ by an element of the stabiliser on the right. Its derivative at the identity coset sends the class of X to T_Xu_0. The kernel of $X \mapsto T_Xu_0$ is the Lie algebra $\mathfrak{h}$ of H. Indeed, a curve in H gives zero derivative of its action on u_0. Conversely, if $T_Xu_0 = 0$, then $\mathcal{R}(\exp(sX))u_0 = \exp(sT_X)u_0 = u_0$ for every s. This one-parameter subgroup lies in H, so $X \in \mathfrak{h}$.

The derivative on G/H is therefore injective. Translation by group elements gives injectivity at every coset. The inverse function theorem in local coordinates gives an immersed image with these tangent spaces. Since G/H is compact, its continuous injective map into the Hausdorff space $\mathbb{R}^N$ is a homeomorphism onto its image. Together with the injective differential, this makes the image embedded. Rank–nullity gives its dimension $\dim \mathfrak{g} - \dim \mathfrak{h}$. Invariance puts every orbit point in $\mathcal{V}$.

Next, differentiate (17.3) in a field direction w:

$$\nabla V(\mathcal{R}(g)u) \cdot \mathcal{R}(g)w = \nabla V(u) \cdot w.$$

Since this holds for every w, orthogonality gives the gradient identity

$$\nabla V(\mathcal{R}(g)u) = \mathcal{R}(g)\nabla V(u). \tag{17.5}$$

At u_0 the right side is zero. Differentiate the left side along $g(s) = \exp(sX)$ at zero. The chain rule gives $D^2V(u_0)T_X u_0 = 0$. This proves the full kernel inclusion, not only a zero quadratic form on tangent vectors.

Finally, the spectral decomposition above identifies the Hessian nullity with the number of massless linearised components. The tangent-space inclusion gives the lower bound. Equality of dimensions holds precisely when these two subspaces coincide. ∎

A *Goldstone mode* is a massless direction forced by this orbit argument. The count concerns independent continuous directions, the quotient $\mathfrak{g}/\mathfrak{h}$. Breaking a discrete symmetry does not by itself give a positive-dimensional tangent space. A Goldstone direction is tangent to a curve of vacua. The straight line with that initial tangent need not stay in the vacuum set.

17.3 The global $O(2)$ model

Take the two real components of Lecture 2, $\Phi = \phi_1 + \mathrm{i}\phi_2$, without a factor $1/\sqrt{2}$. Use the scalar part of the course's Higgs normalisation with $A_{\mathrm{em}} = 0$:

$$L = \frac{1}{2}\partial_\mu \overline{\Phi}\, \partial^\mu \Phi - \frac{\lambda}{8}(|\Phi|^2 - v^2)^2, \qquad \lambda > 0, \quad v > 0. \tag{17.6}$$

This is a global scalar model. Its symmetry is $O(2)$, acting orthogonally on (ϕ_1, ϕ_2), and its vacuum manifold is the circle $|\Phi| = v$. Choose $u_0 = (v, 0)$. The unbroken subgroup consists of I and $\mathrm{diag}(1, -1)$. It has dimension zero, whereas $O(2)$ has dimension one. The orbit lemma therefore forces one massless direction.

Figure 17.1 separates the radial direction from the tangent to the circle at the chosen vacuum. The tangent vector is an infinitesimal variation; the full orbit follows the circle.

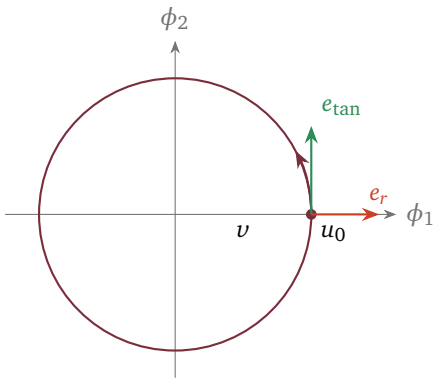


Figure 17.1: The vacuum circle and the chosen point $u_0 = (v, 0)$. The orbit tangent is vertical there; the radial direction is horizontal.

The gradient and Hessian of the potential, writing $u = (\phi_1, \phi_2)$, are

$$\nabla V(u) = \frac{\lambda}{2}(|u|^2 - v^2)u, \qquad D^2V(u) = \frac{\lambda}{2}\big[(|u|^2 - v^2)I + 2uu^{\mathsf{T}}\big].$$

At the chosen vacuum the mass matrix and tangent are

$$\mathsf{M} = \begin{pmatrix} \lambda v^2 & 0 \\ 0 & 0 \end{pmatrix}, \qquad Tu_0 = \begin{pmatrix} 0 \\ v \end{pmatrix}, \qquad T = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \tag{17.7}$$

There is one radial mass $m_h = \sqrt{\lambda}\, v$ and one massless tangential mode. The Hessian kernel is exactly the orbit tangent.

The phase coordinate requires its own kinetic normalisation. Locally near u_0, with $|h| < v$ and phase ϑ near zero, write $\Phi = (v + h)e^{i\vartheta}$. The density becomes

$$L = \frac{1}{2}\partial_\mu h \partial^\mu h + \frac{1}{2}(v + h)^2 \partial_\mu \vartheta \partial^\mu \vartheta - \frac{\lambda}{8}(2vh + h^2)^2.$$

Keeping terms of total degree two in the perturbations and their derivatives gives the quadratic density. With the canonically normalised phase field $\xi = v\vartheta$, it is

$$L^{(2)} = \frac{1}{2}\partial_\mu h \partial^\mu h + \frac{1}{2}\partial_\mu \xi \partial^\mu \xi - \frac{1}{2}\lambda v^2 h^2. \tag{17.8}$$

Its equations are $(\Box + \lambda v^2)h = 0$ and $\Box \xi = 0$. The radial potential starts with $\frac{1}{2}\lambda v^2 h^2$ because $(\lambda/8)(2vh + h^2)^2$ has that quadratic coefficient. Changing $\lambda/8$ to $\lambda/4$ would change the radial squared mass by a factor of two.

17.4 An accidental zero curvature

A zero Hessian eigenvalue need not be a Goldstone direction. Take the one-component potential $V(s) = s^4$ and the trivial compact group $G = \{e\}$. Its unique vacuum is $s = 0$ and its orbit is a point, but

$$V'(0) = 0, \qquad V''(0) = 0, \qquad \ker D^2V(0) = \mathbb{R}.$$

The linearised equation is massless even though no continuous symmetry is broken. The full restoring force is $V'(s) = 4s^3$, which is nonzero for $s \neq 0$. There is no nonconstant curve of vacua through zero. The potential is not flat in a neighbourhood of its minimum.

Thus orbit directions give a lower bound, not a complete classification of Hessian zeros without the kernel-equality hypothesis. Lecture 21 must retain this distinction when it couples the phase to a gauge field. Lecture 18 first studies perturbations about a nonconstant kink.

Summary. For a C^2 invariant potential, a compact Lie group with a smooth linear orthogonal action gives a vacuum orbit G/H. The orbit has dimension $\dim G - \dim H$, and every tangent lies in the Hessian kernel. With canonical kinetic terms this forces at least that many massless linearised modes. The count is exact when the kernel has no additional directions. The global $O(2)$ model with potential $(\lambda/8)(|\Phi|^2 - v^2)^2$ has radial mass $\sqrt{\lambda}\, v$ and one canonically normalised massless phase. A vanishing Hessian alone does not establish a broken symmetry or a continuum of vacua.

Self-check. Lecture 15 has $V(s) = \frac{1}{2}(1 - s^2)^2$ and the discrete symmetry $s \mapsto -s$. At the vacuum $s = 1$, must there be a Goldstone mode? Compute the mass.

Answer. The orbit has two points and dimension zero, so the lemma forces no massless mode. The curvature is $V''(1) = 4$, giving mass two.

Self-check. At the rotated $O(2)$ vacuum $u_\theta = v(\cos\theta, \sin\theta)$, identify the radial and tangential eigenvectors and their squared masses.

Answer. The radial unit vector is $(\cos\theta, \sin\theta)$, with squared mass λv^2. The tangent unit vector is $(-\sin\theta, \cos\theta)$, with squared mass zero. The Hessian is $\lambda u_\theta u_\theta^{\mathsf{T}}$.

Self-check. Why is $\xi = v\vartheta$, rather than ϑ, the canonically normalised phase variable? Does this rescaling give it a mass?

Answer. The phase kinetic term at the vacuum is $\frac{1}{2}v^2 \partial_\mu \vartheta \partial^\mu \vartheta$. The rescaling makes its coefficient $1/2$. There is still no quadratic potential for ξ, so it remains massless.

Self-check. Which part of the orbit–Hessian argument needs a minimum, rather than only a critical point? What does $V(s) = s^4$ show about the converse?

Answer. Gradient covariance gives the kernel inclusion at any critical point. The minimum makes the Hessian positive semidefinite (nonnegative squared masses) and puts the orbit in the vacuum set; at a mere critical point the orbit consists of critical points. The quartic example has an additional zero eigenvalue without a symmetry orbit direction, so a zero Hessian does not imply symmetry breaking.